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EDUCATION

- Ph.D. student, School of Advanced Materials Science & Engineering (AMSE),** Mar. 2020 – Present
Sungkyunkwan University (SKKU), Graduate School Suwon, South Korea
Advisor: Prof. Jung Heon Lee (email: jhlee7@skku.edu)
- B.S., School of Advanced Materials Science & Engineering, SKKU** Mar. 2014 – Feb. 2020
Suwon, South Korea

RESEARCH INTERESTS

Nanoparticle, Self-assembly, Ceramics, Biomaterials

PUBLICATIONS

First and Co-first author publications ([†]*Equal contribution*)

- [7] **C. H. Park[†]**, P. R. Sundharbaabu[†], Yoon Sung Cho[†], J. H. Heo*, J. H. Kang*, J. H. Lee*, “Engineering "Plug-and-Play" Functional Metallized Ceramics Using DNA as Synthesis Scaffolds” (*To be submitted*)
- [6] J. Cho[†], **C. H. Park[†]**, P. R. Sundharbaabu[†], S.-U. Ju, D. Shin, J. H. Heo*, M. Kim*, J. H. Lee*, “Transformative Mechanical and Piezoelectric Properties in Organic–Inorganic Hybrid Materials for Biomedical Applications” *ACS Nano* **19** (41), 36035–36081 (2025).
- [5] **C. H. Park[†]**, H. Song[†], D.-H. Kim, J. Kim, S. H. Jeong, J. B. B. Rayappan, J. H. Heo, J. H. Lee, “Functional ATP-Based Fabric Organic Dyes with Flame Retardancy and Antibacterial Activity” *Chemical Engineering Journal* **506**, 159912 (2024).
- [4] J. W. Lee[†], B. Lee[†], **C. H. Park[†]**, J. H. Heo, T. Y. Lee, D. Lee, J. Bae, P. R. Sundharbaabu, W. K. Yeom, S. Chae, J.-H. Lim, S.-W. Lee, J.-S. Choi, H.-B. Bae, J.-Y. Choi, E.-H. Lee, D. S. Yoon, G. Y. Yeom, H. Shin, J. H. Lee, “Monolithic DNApatite: An Elastic Apatite with Sub-Nanometer Scale Organo–Inorganic Structures” *Advanced Materials* **36** (41), 2406179 (2024).
- [3] H. Song[†], **C. H. Park[†]**, S. H. Jeong, J. H. Heo, J. H. Lee, “Synergistic adenosine triphosphate/chitosan bio-coatings on polyurethane foam for simultaneously improved flame retardancy and smoke suppression” *ACS applied Polymer Materials* **5** (6), 4388–4399 (2023).
- [2] S. H. Jeong[†], **C. H. Park[†]**, H. Song, J. H. Heo, J. H. Lee, “Biomolecules as green flame retardants: recent progress, challenges, and opportunities” *Journal of Cleaner Production* **368**, 133241 (2022).
- [1] S. H. Kim[†], **C. H. Park[†]**, J. H. Heo, J. H. Lee, “Progress and perspectives of metal-ion-substituted hydroxyapatite for bone tissue engineering: comparison with hydroxyapatite” *Journal of Korean Ceramic Society* **59** (3) (2022).

Co-author publications

- [4] H. Song[†], S. H. Jeong[†], **C. H. Park**, M. J. Kim, H. Kim, J. H. Heo, J. H. Lee, “A multifaceted heat-switching nanofiller: Design of heat dissipation–insulation switching materials above a critical temperature” *Chemical Engineering Journal* **497**, 154406 (2024).
- [3] D.-H. Kim[†], J. Bae[†], J. H. Heo, **C. H. Park**, E. B. Kim, J. H. Lee, “Nanoparticles as next-generation tooth-whitening agents: Progress and perspectives” *ACS Nano* **16** (7), 10042–10065 (2022).
- [2] C. Kim[†], J. W. Lee[†], J. H. Heo, **C. H. Park**, D.-H. Kim, G. S. Yi, H. C. Kang, H. S. Jung, H. Shin, J. H. Lee, “Natural bone-mimicking nanopore-incorporated hydroxyapatite scaffolds for enhanced bone tissue regeneration” *Biomaterials Research* **26** (1), 7 (2022).
- [1] S. H. Jeong[†], J. H. Heo, J. W. Lee, M. J. Kim, **C. H. Park**, J. H. Lee, “Bioinspired adenosine triphosphate as an “all-in-one” green flame retardant via extremely intumescent char formation” *ACS Applied Materials & Interfaces* **13** (19), 22935–22945 (2021).

CONFERENCE

Contributed Talks

- [5] **C. H. Park**, J. W. Lee, B. Lee, D. Lee, J. H. H, T.-Y. Lee, J. Bae, P. R. Sundharbaabu, J. H. Lee*, “Monolithic assembly of elastic apatite with sub-nanometer scale integration of dual-phase organo-inorganic structures” Oral presentation, *ACS Fall 2025*, Washington DC, USA. August 17–21.
- [4] **C. H. Park**, J. W. Lee, B. Lee, D. Lee, J.-H. Lim, S.-W. Lee, P. R. Sundharbaabu, J. Bae, D. S. Yoon, E.-H. Lee, J. H. Heo, J. H. Lee, “DNA-Assembled Elastic Apatite Monoliths with Sub-Nanometer Precision”, Oral presentation, *The Korean Ceramic Society Spring 2025*, Yeosu, Korea. April 16–18.
- [3] **C. H. Park**, J. Bae, J. W. Lee, S. Chae, S. Oh, S. H. Kim, J.-Y. Choi, J. H. Lee*, “Inorganic Molecular Nanowire-Based Extracellular Matrix-Mimicking Scaffolds for Tissue Engineering”, Oral presentation, *The 7th International Conference on Electronic Materials and Nanotechnology for Green Environment (ENGE 2022)*, Jeju, Korea. November 6–9.
- [2] **C. H. Park**, S. H. Jeong, J. W. Lee, M. J. Kim, J. H. Heo, J. H. Lee*, “Biomolecular Adenosine Triphosphate as One Compacted Green Flame Retardant through Char formation with extreme intumescence”, Oral presentation, *The Korean Ceramic Society Fall 2021*, Jeju, Korea. November 3–5.
- [1] **C. H. Park**, C. Kim, J. W. Lee, J. H. Heo*, D.-H. Kim, G. S. Yi, H. C. Kang, H. S. Jung*, H. Shin*, J. H. Lee*, “Fabrication of Natural Bone Mimetic Hydroxyapatite-Based Scaffold Containing Nanopores for the Improvement of Bone Regeneration”, Oral Presentation, *The Korean Institute of Metals and Materials Spring 2021*, Hoengseong, Korea. April 28–30.

Poster Sessions

- [7] **C. H. Park**, J. W. Lee, B. Lee, D. Lee, J.-H. Lim, S.-W. Lee, P. R. Sundharbaabu, J. Bae, D. Shin, D. S. Yoon, E.-H. Lee, J. H. Heo*, J. H. Lee*, “Enhancing the Mechanical Properties of Apatite: A DNA-Driven Approach” Poster presentation (online), *Royal Society of Chemistry’s 2025 #RSCPoster Conference* on LinkedIn.
- [6] **C. H. Park**, J. W. Lee, B. Lee, D. Lee, J.-H. Lim, S.-W. Lee, P. R. Sundharbaabu, J. Bae, D. S. Yoon, E.-H. Lee, J. H. Heo, J. H. Lee, “DNA-Based Monolithic Assembly of Elastic Apatite with Dual Organo-Inorganic Structuring at the Sub-Nanometer Scale”, Poster presentation, *The Korean BioChip Society Spring 2024*, Jeju, Korea. November 13–15.
- [5] **C. H. Park**, J. W. Lee, B. Lee, D. Lee, J.-H. Lim, S.-W. Lee, P. R. Sundharbaabu, J. Bae, D. S. Yoon, E.-H. Lee, J. H. Heo, J. H. Lee, “Elastic DNApatite with Sub-Nanometer Scale Nacre-Mimetic Structures”, Poster presentation, *The Korean Ceramic Society Fall 2024*, Seoul, Korea. October 16–18.
- [4] **C. H. Park**, H. Song, S. H. Jeong, J. H. Heo*, J. H. Lee*, “Sustainable Cotton Fabric Treatment Using Adenosine Triphosphate dye for Enhancing Flame Retardancy and Color Retention”, Poster presentation, *The Korean BioChip Society Spring 2023*, Yeosu, Korea. May 31–June 2.
- [3] **C. H. Park**, J. Bae, J. W. Lee, S. Chae, S. Oh, S. H. Kim, J.-Y. Choi, J. H. Lee*, “Molecular Chain Based on Inorganic Nanowire as Extracellular-Matrix Mimicking Scaffold with Bio-essential Properties”, Poster presentation, *The Korean BioChip Society Fall 2022*, Jeju, Korea. November 2–4.
- [2] **C. H. Park**, S. Kannappan, D.-H. Kim, S. H. Jeong, J. Bae, E. B. Kim, J. W. Lee*, J. H. Lee*, “Biomimetic Inorganic Molecular Nanowires as a Muscle Extracellular-Matrix Scaffold for Enhanced Bioactivity”, Poster presentation (online), *The Korean BioChip Society Fall 2021*, Korea. November 17–19.

- [1] **C. H. Park**, J. W. Lee, S. Chae, S. Oh, S. H. Kim, K. H. Choi, J. H. Heo, J.-Y. Choi*, J. H. Lee*, “Single-Chain Atomic Crystals as Extracellular Matrix-Mimicking Material with Exceptional Biocompatibility and Bioactivity”, Poster presentation, *The Korean Ceramic Society Fall 2020*, Daejeon, Korea. November 23–25.

SCHOLARSHIPS & AWARDS

Academic Excellence Award , 8 th Brain Korea 21 Outstanding Graduate Student Awards in AMSE, SKKU	2025
Best Oral Presentation Award , Conference of The Korean Ceramic Society Spring (KCerS)	2025
Best Poster Presentation Award , Conference of The Korean Ceramic Society Fall (KCerS)	2024
Best Presentation Award , Research Matters Plus, SKKU	2023
Academic Excellence Scholarship , - Half-tuition scholarship for the duration of the graduate program in AMSE, SKKU	2020–2023
Brain Korea 21 Scholarship , National Research Foundation of Korea (NRF)	2020–Present
Samsung Dream Class Scholarship , Samsung Welfare Foundation	2019

PATENTS

- [6] J. H. Lee, J. H. Heo, S. H. Jeong, **C. H. Park**, H. Song, A Flame-Retardant Plastic Based on Adenosine Triphosphate and Epoxy Resin (South Korea – Application No. 10-2021-0101426; Registration No. 10-2559919).
- [5] J. H. Lee, J. H. Heo, **C. H. Park**, S. H. Jeong, H. Song, A Flame-Retardant Plastic Based on Adenosine Triphosphate and Polylactic Acid (South Korea – Application No. 10-2021-0101871; Registration No. 10-2632538).
- [4] J. H. Lee, J. H. Heo, **C. H. Park**, S. H. Jeong, H. Song, A Flame-Retardant Dye Using Adenosine Triphosphate as an Eco-Friendly Phosphorus-Based Flame-Retardant Additive (South Korea – Application No. 10-2021-0102362; Registration No. 10-2494785).
- [3] J. H. Lee, J. H. Heo, **C. H. Park**, S. H. Jeong, H. Song, A Method for Enhancing the Flame Retardancy of Polypropylene Resin with Adenosine Triphosphate (South Korea – Application No. 10-2021-0101463; Registration No. 10-2559916).
- [2] J. H. Lee, J. W. Lee, **C. H. Park**, J. H. Heo, T.-Y. Lee, D. H. Kim, An Organic-Inorganic Hybrid Monolithic Compound Using Nucleic Acid Monomers (ADP, AMP, TMP, GMP, CMP) (South Korea – Application No. 10-2021-0000787; Registration No. 10-2525363).
- [1] J. H. Lee, J. W. Lee, T.-Y. Lee, J. H. Heo, **C. H. Park**, D. H. Kim, An Organic-Inorganic Hybrid Monolithic Compound Using Phosphorylated Amino Acids (Phosphoserine, Phosphotyrosine, Phosphothreonine) (South Korea – Application No. 10-2021-0000751; Registration No. 10-2525360), Technology transferred (Recipient: CG Bio incorporation) Dec. 2022.

RESEARCH EXPERIENCE

Graduate Student Researcher | Biological & Nanoscale Materials Lab (BNML), SKKU

Ceramic-supported Functional Nanomaterials Dec. 2024 – present

Co-principal investigator: Prof. Jung Heon Lee (SKKU) & Prof. Joohoon Kang (Yonsei Univ.).

- Contributed to the development of a material exhibiting high chemical resistance and catalytic performance through the sequential process of metal-catalyzed reaction and crystallization on a DNA scaffold.
- *Manuscript in preparation.*

CHIRON: Ceramic Hybrid Interfaces foR Optimal cell stimulation Dec. 2024 – present

Co-principal investigator: Prof. Jung Heon Lee (SKKU) & Prof. Tobias Fey (FAU).

Funded by the National Research Foundation of Korea (RS-2024-00438009)

- **The conceptual designing** of an international joint research program (Korea-Germany), **defining** key research objectives and **formulating** experimental strategies for developing advanced ceramic-hybrid interfaces for optimal cell stimulation.

Development of High-Toughness Elastoceramic Materials

Mar. 2020 – Feb. 2025

Co-principal investigator: Prof. Jung Heon Lee (SKKU) & Dae Sung Yoon (Korea Univ.) & Eun-Ho Lee (SKKU)

Funded by the National Research Foundation of Korea (NRF-2020R1A2C2006100).

- Engineered a novel elastoceramic material with an organic-inorganic hybrid structure by developing a DNA-backbone synthesis method. **Characterized its unique mechanical properties using nanoindentation and UTM, and confirmed its structure via TEM and XRD**, leading to a co-first author publication in *Advanced Materials*.
- Publication: *Advanced Materials*, 36(41), 2406179 (2024).

Development of Hydroxyapatite Based Artificial Bone Tissue Generation

Mar. 2020 – Dec. 2022

Principal investigator: Prof. Jung Heon Lee (SKKU)

Funded by the National Research Foundation of Korea (NRF-2018M3C1B7021997).

- **Led in vitro validation**, leading experiments with MC3T3-E1 pre-osteoblast cells to assess cytotoxicity, proliferation, and biocompatibility, directly.
- Publications: *Biomaterials Research*, 26(1), 7 (2022); *Journal of the Korean Ceramic Society*, 59(3) (2022).

Functional Nanomaterials for Thermal Management & Flame Retardancy

Mar. 2021 – Jan. 2025

Principal investigator: Prof. Jung Heon Lee (SKKU)

- **Developed a portfolio** of sustainable flame retardants from biomolecules (ATP), creating multifunctional dyes, polymer coatings, and nanofillers that significantly enhanced fire safety and smoke suppression in cotton fabrics, commercial polymers, and batteries, respectively.
- Publications: *Chemical Engineering Journal*, 506, 159912 (2025); *Chemical Engineering Journal*, 497, 154406 (2024); *ACS Applied Polymer Materials*, 5(6) (2023); *Journal of Cleaner Production*, 368 (2022).

TEACHING EXPERIENCE

Mentoring junior students in BNML

Davin Shin (PhD student)

Feb. 2024 – present

Research mentor, Undergraduate Research Program (URP), SKKU

Mar. 2022 – Aug. 2022

- Mentored an undergraduate student project focused on the synthesis of biochar-hydrogel composites for water purification.
- Provided hands-on training and guidance on experimental procedures and data analysis.

Teaching assistant

Mar. 2021 – Aug. 2022

- Instructed and mentored undergraduate students on experimental procedures and safe instrument operation.
- Graded lab reports and examinations, providing constructive feedback to enhance student learning.

SKILLS

Materials characterization

- Microscopy: Scanning Electron Microscopy (SEM), Transmission Electron Microscopy (TEM)
- Spectroscopy: X-ray Diffraction (XRD), X-ray Photoelectron Spectroscopy (XPS), FT-IR, UV-Vis, Circular Dichroism (CD)
- Particle Analysis: Dynamic Light Scattering (DLS), Zeta Potential
- Mechanical Properties: Nanoindentation, Universal Testing Machine (UTM) for fracture mechanics analysis

Biological assays & cell culture

- Cell Lines: MC3T3-E1 (pre-osteoblast), L929 (fibroblast)
- Techniques: Aseptic cell culture and maintenance, cytotoxicity assays (e.g., MTT, WST), cell proliferation analysis, fluorescence microscopy (Live/Dead staining).

REFERENCES

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